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Course and Section: CSIS 4495-002

Power Quality Analyzer - Midcourse Check-in

GitHub Repo: F26_4495_S2_OphiliaF.git

Midcourse Progress Video:
https://drive.google.com/file/d/1Xqom9R8Akl91LX0TbpZJeIAKPZK30V1W/view?usp=sharing

Introduction

The web app is an individual applied research project that aims to classify an uploaded voltage recording as normal, voltage sag, voltage swell, interruption, or harmonic distortion. This report describes the working interface and signal examples completed so far, explains changes to the original proposal, and presents the remaining implementation and research plan. The current prototype can load a cleaned CSV recording, display its waveform, calculate basic measurements, add measurement noise, and export the displayed signal. A Flask server serves the interface locally. The trained classifiers and classification API have not been implemented. Consequently, the current application demonstrates signal preparation and exploration but does not yet produce a machine-learning diagnosis.

Domain and practical problem

Power quality concerns the characteristics of an electrical supply. In this project, a sag is a temporary reduction in voltage magnitude, a swell is an increase, an interruption is a loss or near loss of supply, and harmonics alter the shape of a regular sine wave. Normal operation is included as a reference class. The examples are simplified mathematical signals rather than measurements from electrical equipment.

A possible application is an offline maintenance workflow. A technician investigating an equipment shutdown could export an existing voltage recording, prepare it in the supported format, and inspect it in the browser. Once the classifier is implemented and evaluated, the application could provide a candidate disturbance label for further investigation. No industrial partner deployment, sensor connection, or equipment protection function has been implemented.

Research questions and assumptions

My project is mostly built upon the research paper I found. I am trying to incorporate the findings into the web applications. Here are the assumptions:

1. The research asks whether Random Forest and Support Vector Machine classifiers can distinguish the five selected classes, and how their performance changes when Gaussian measurement noise is added.
2. A related comparison asks whether frequency features retain more useful class information than time features under the selected noise conditions. [1] Review machine learning and deep learning approaches to power quality disturbance classification, including the role of signal processing in feature extraction. Their case study reports 96.6% testing accuracy for Bagged Trees.
3. Abu Nassar and Morsi [2] investigate how cyberattacks can be distinguished from disturbances and normal operation in smart digital substations. They use continuous wavelet representations and a convolutional neural network. Scope of the research contribution: The intended contribution is a small, reproducible comparison using the same underlying test examples across clean, 40 dB, 30 dB, and 20 dB conditions.

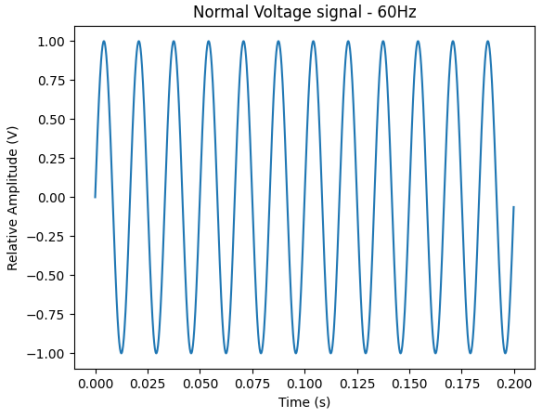
The proposal scheduled a full synthetic dataset and exploratory analysis before Progress Report 1, followed by feature extraction, clean-signal model training, and a basic connected application for the midcourse milestone. Those research milestones have not been met. The implemented work is concentrated in illustrative signal generation and the browser interface, with a local Flask serving layer added on.

Domain Knowledge refresher: (The code is in the normal_signal.py file)

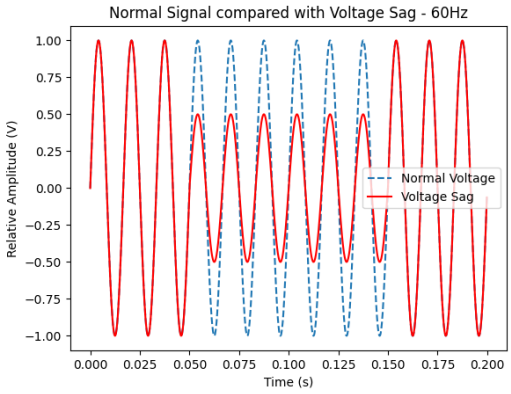
On October 18, I worked through a Python learning script using NumPy and Matplotlib to generate and inspect normal, sag, swell, interruption, and harmonic examples. It starts with a 60 Hz sine wave, a sample
rate of 6,000 measurements per second, a duration of 0.2 seconds, and a relative peak amplitude of 1.0. These settings produce 1,200 samples.
The base formula is v(t) = A sin(2 pi f t + phase). The script constructs each timestamp by dividing an integer sample index by the sample rate. This gives a regular interval of 1/6000 seconds. Amplitude is relative rather than calibrated physical volts

The output:

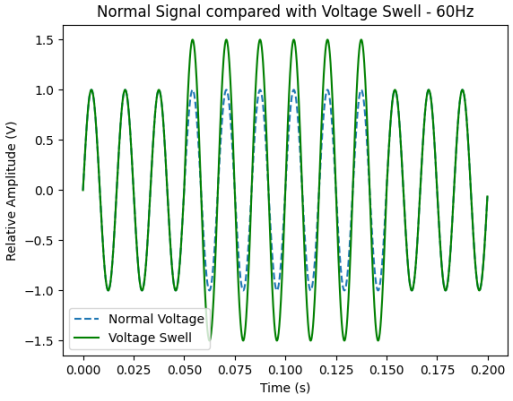
Normal Voltage wave



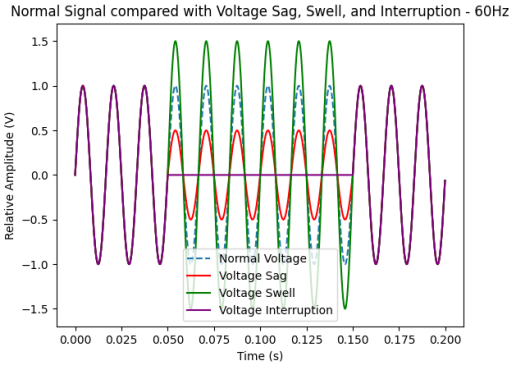
Voltage Sag:



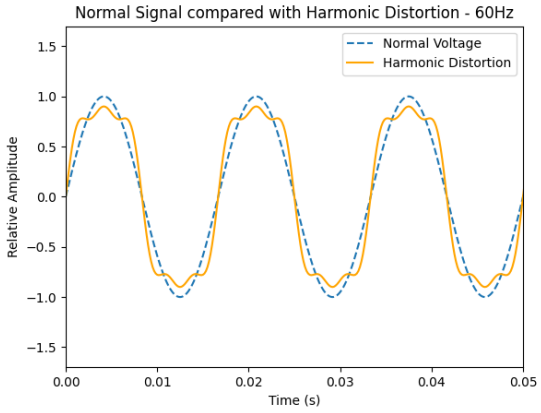
Voltage Swell



Voltage Interruption

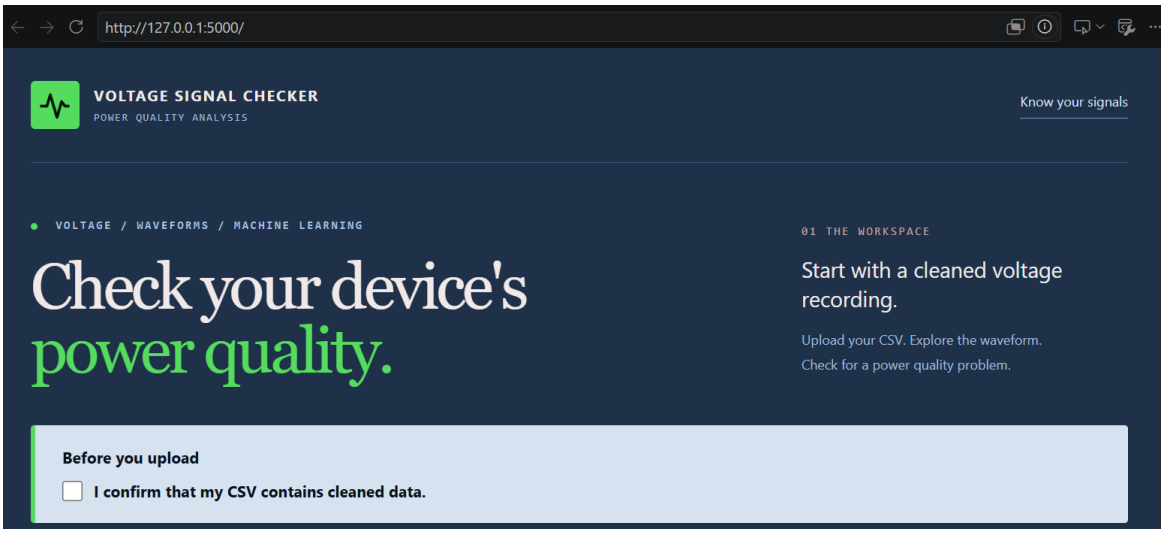


Harmonic Distortion

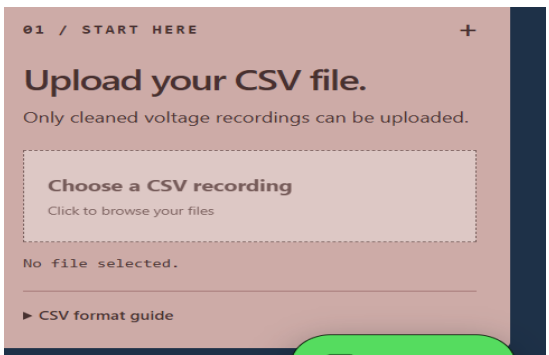


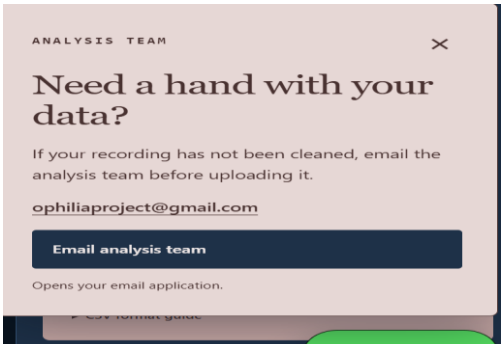
Frontend:

The upload workflow starts with a declaration that the recording has been cleaned.

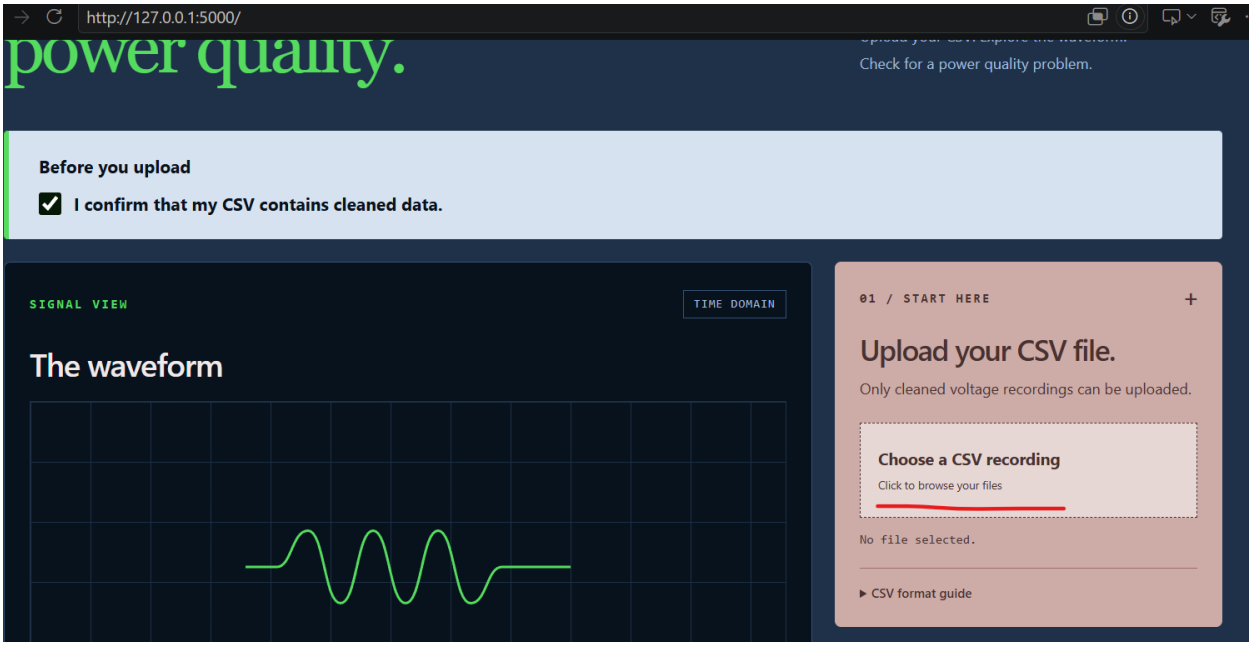


The reason why I added a checkbox is to make sure that the technician or anyone who use this application is only uploading the cleaned data. If the data is not cleaned, as of now, I ask them to reach out to me via email. I incorporated a chat bubble (Not a chatbot system) to show the email to which they should send the CSV files to be cleaned.





After clicking on the checkbox confirming that the dataset is clean, only then does the option to upload the CSV file become active. This is done using JavaScript.

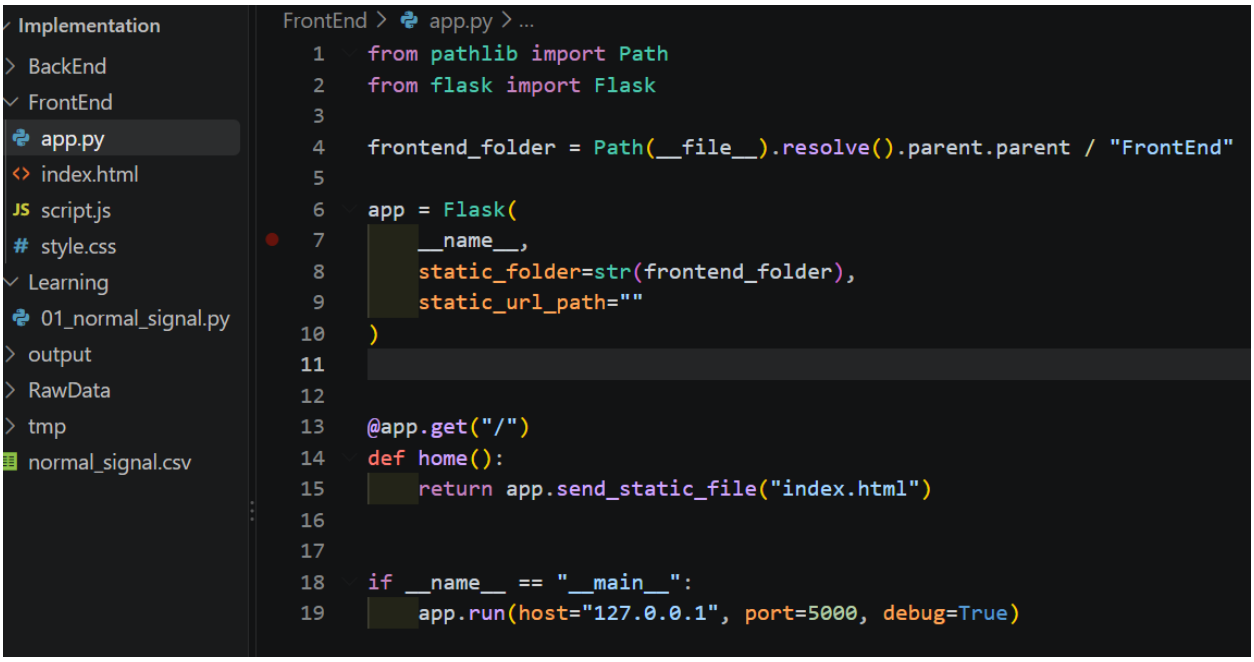




After a valid CSV is loaded, the canvas displays voltage values against time. The dashboard shows the number of measurements, an estimated sample rate, and the RMS of the displayed signal. The original recording remains available while a separate preview array holds the version with added noise.

Figure 5. Actual browser output for the project sample: 1,200 measurements, 6,000.0 Hz estimated sample rate, and 0.7071 relative
RMS. For this complete-cycle sine-wave example, the RMS is approximately 1 divided by the square root of 2.

Backend:
I used the Flask framework for the backend.



Summary of the Initially Proposed Research Project

My original proposal described a web application that would use machine learning to identify power quality disturbances in uploaded voltage recordings. The application was intended to make the analysis accessible through a browser, allowing users to inspect a recording and receive a predicted signal category.

The proposed research would compare two supervised machine learning classifiers: Random Forest and Support Vector Machine. Before training, each signal would be converted into numerical features describing its characteristics. These would include time-domain measurements, such as RMS and peak amplitude, and frequency-domain features obtained using the Fast Fourier Transform. The comparison would investigate how the selected features and classifiers affect classification performance.

Changes to the Proposal and Justification

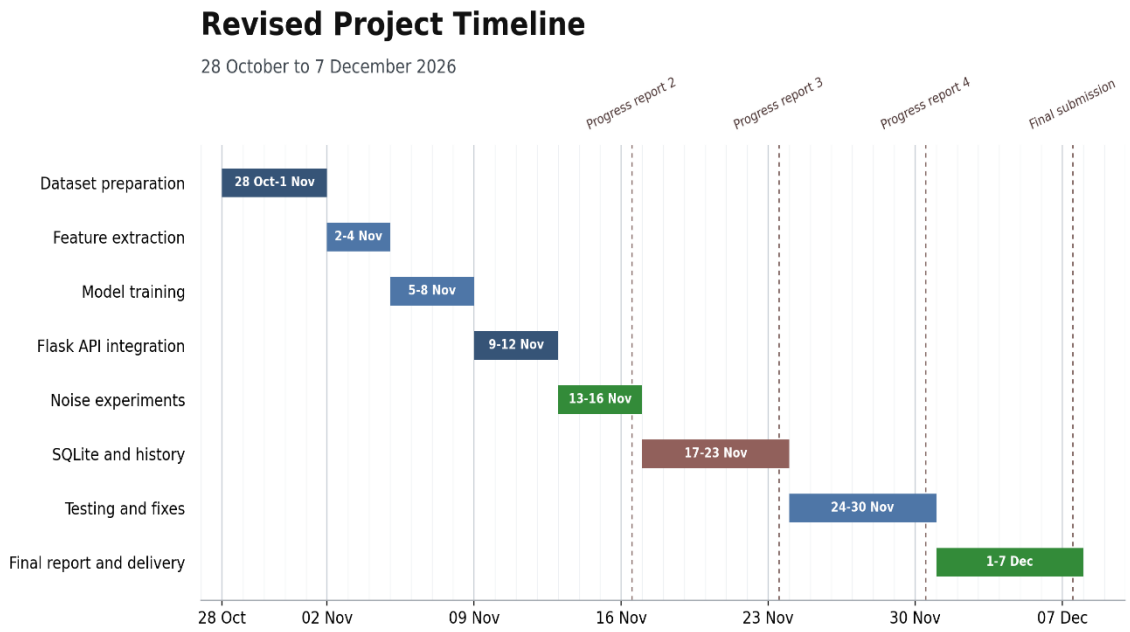
The main project idea remains unchanged from my original proposal. It still focuses on classifying five voltage signal categories using Random Forest and SVM and evaluating their performance under different noise levels. The changes concern interface details and the implementation schedule, while the research objectives remain the same.

Project Planning and Timeline

Feature extraction, Random Forest and SVM training, and the classification API remain unfinished. The implementation sequence has therefore changed: I first developed the signal examples and browser workflow, and the remaining work will focus on completing and evaluating the machine learning pipeline. SQLite storage and the history panel will follow once classification is working.

Next Steps:

1. I must work on the machine learning part of the application and must write the code to analyze the signals when the noise is added.
2. I must add the database to save the results that were evaluated in the past.
3. After completing the application, I plan to test the Flask API using Postman. Application hosting will be considered separately. (I must revise these concepts; I did this years ago)



Planned dates	Task	Expected deliverable
Oct 28–Nov 1	Prepare dataset	Labelled examples of all five signal classes.
Nov 2–4	Extract features	Validated time-domain and FFT features.
Nov 5–8	Train models	Trained Random Forest and SVM models with evaluation results.
Nov 9–12	Integrate Flask API	Working upload-to-prediction workflow.
Nov 13–16	Test noise robustness	Results for clean, 40, 30, and 20 dB signals; Progress Report 2.

Planned dates	Task	Expected deliverable
Nov 17–23	Add SQLite and history	Saved analysis records and history view; Progress Report 3.
Nov 24–30	Test application	Verified features, bug fixes, and installation steps; Progress Report 4.
Dec 1–7	Finalize project	Final report, updated README, working application, and demo materials.

AI Use Section:

AI T(I must Account Type	**Specific feature for which the AI tool was used**	**Value Addition** ***What value did you add over and above what AI did for you?***	
Claude	Pro	Engineering terminology and discussion of dataset access.	It helped me refresh the domain knowledge
Chatgpt	Free	Helped me with writing the HTML and CSS code	I know the basics of HTML, CSS and JS, so I gave specific instructions to complete the Frontend

- ***Appendix:*** *Prompt history is in Appendices A and B*

Work log

Date	Hours	What was done
9-Sep-26	2.5	Brainstormed project ideas during the first lecture, discussed them with the professor, and received approval to proceed with the project.
16-Sep-26	0.5	Spoke with an engineering manager about becoming my industry partner. However, their area of expertise is different from the focus of my project.
18-Sep-26	1	Met with Kalachinskaya, Olga, Faculty Subject Librarian for ENGR, MATH, CSIS, CMPT, VTEC, and VOAC, to discuss how to find research papers to support my Applied Research Project.
18-Sep-26	0.25	Created GitHub repo (F26_4495_S2_OphiliaF); added professor kandhadaip@douglascollege.ca as collaborator
20-Sep-26	3	Found three research papers that can be used to support my project.
23-Sep-26	1	In-person check-in with professor; confirmed project topic and scope
25-Sep-26	4	Wrote and edited the project proposal, consolidated the content, and simplified the wording to make the proposal clearer and easier to understand.
26-Sep-26	0.25	Meeting with Professor
28-Sep-26	1	Finalizing the proposal and submitting it
30-Sep-26	2	Spend some time on the DataCamp website to revise the concept of EDA
7-Oct-26	0.5	Wrote a few lines of HTML code
8-Oct-26	1	Revised Git commands and organized the GitHub repo
11-Oct-26	2	Found the dataset. IEEE dataset was a paid one; hence, I moved to Kaggle and found the dataset.
13-Oct-26	1	Worked on the Frontend of the web app
18-Oct-26	2	Wrote the code to generate and plot five synthetic voltage signal types using NumPy and Matplotlib. Exported the normal signal to CSV
19-Oct-26	1	Worked on the Frontend of the web app (HTML part)
20-Oct-26	2	Worked on the Frontend of the web app (HTML and JS part)
21-Oct-26	0.5	Implemented Flask (Backend) and changed some CSS code
22-Oct-26	0.5	Changed the front end (worked on CSS mostly)
27-Oct-26	1	Started working on the mid-course check (Missed deadline)
28-Oct-26	4	Recorded the video and worked on the report, worked on the Report and uploaded it to GitHub (In-class work)

Contribution evidence and record keeping

Git records show frontend check-in on October 19, backend and interface changes on October 27, and interface cleanup on October 28. I haven't been more consistent, but I will be going forward.

Closing and References

The completed midcourse work establishes an interactive signal exploration interface, mathematical examples of the five selected signal classes, and a local Flask serving structure. The present application can demonstrate upload, plotting, basic measurements, noise preview, CSV download, and contact-panel behavior. It cannot yet demonstrate a learned classification result.

Acknowledgements

I acknowledge the course instructor for project guidance and faculty subject librarian Olga Kalachinskaya for support in finding research literature. AI assistance is disclosed in the sections and the prompt appendices. I did use AI for creating the CSS and JS codes., also altered a small portion of HTML code.

References

[1] S. Jain, A. Satsangi, R. Kumar, D. Panwar, and M. Amir, Intelligent assessment of power quality disturbances: A comprehensive

review on machine learning and deep learning solutions, Computers and Electrical Engineering, vol. 123, article 110275, 2025.

https://doi.org/10.1016/j.compeleceng.2025.110275

[2] A. Abu Nassar and W. G. Morsi, Detection of Cyber-Attacks and Power Disturbances in Smart Digital Substations using Continuous

Wavelet Transform and Convolution Neural Networks, Electric Power Systems Research, vol. 229, article 110157, 2024.

https://doi.org/10.1016/j.epsr.2024.110157

[3] S. Zhao, F. Blaabjerg, and H. Wang, An Overview of Artificial Intelligence Applications for Power Electronics, IEEE Transactions on

Power Electronics, vol. 36, no. 4, pp. 4633-4658, 2021. https://doi.org/10.1109/TPEL.2020.3024914

Appendix A

Prior recorded proposal prompts

These entries are reproduced from the prompt-history section of the submitted proposal. They do not replace a full export of other AI sessions used later in the project.

A1 September 9 2026

I am a student who is new to research. Where should I start? How do I write a research proposal?

A2 September 18 2026

Is it feasible to research cyberattack detection in smart grids within one semester?

A3 September 20 2026

Is there any application of this research in BC Hydro or TransLink?

A4 September 23 2026

Can you show me what the IEEE DataPort dataset looks like?

A5 September 24 2026

I want to create something realistic that will improve my SQL skills and data analysis skills.

A6 September 24 2026

Can I create an app or chatbot system for this? My professor wants a product that incorporates front-end and back-end, a full-stack operation project.

A7 September 25 2026

Give me all the prompts I asked for, with dates.

Appendix B:

October 18 2026 1414

i wass css, java files spetarte from htmll

October 19 2026 1015

give me the code to paste in html and then css, later js

October 19 2026 1017

add script.js

October 27 2026 2317
i don't like the index.html interace, , can you make the background #1E3148
#9E6F6D
#E6D7D5
#43B14B
#D7E2F0
#F0E9E8
#E6D7D5
#ABFAAB
#EDEFF0
#A2BDDD
#D4BEBC
#CDABA7
#55DC5F
#C2C9CC
#6E98CB
#BB9391
#B67F79
#43B14B
#9BA3A8
#4E76A7
#9E6F6D
#91605B
#338B39
#7B8285
#365477
#73504F
#66433E
#226327
#595F61
#1E3148
#483130
#3B2522
#113C14
#383B3D
#08121D
#211414
#150A09

October 27 2026 2305

can you make the changes in the github and the vsc at the same time?

Appendix C Submission checklist

Checklist (CHECK EACH ENTRY TO VERIFY THAT IT HAS BEEN DONE)

☒	**GitHub Repository Code Check-In**
☒	**Video Demo of the Product**
☒	**Video Link:**
☒	**Installation Instructions:**
☐	**MidCourse Progress Report Blackboard Submission:** Completed Report submitted on Blackboard by the Team Lead
☒	**GitHub Repository MidCourse Progress Report Check-in:**